

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session e34d7fc4 · Data hash 9524...159d

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 22:11 and 22:26 UTC. Reported results: test_rmse_mean 0.40378, paper_test_rmse_mean 0.403006. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	e34d7fc4-c4e3-4b97-9b94-87badf99b360
Initialized	2026-10-07T22:11:27Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	6dbb1edd49cf5b8d7bd0d03b6285b0cf3b4c8d8bb90e3936161b82476c5cbc3f
Disclosure	open
Source code, run 1	a612d4ec0787bdf6b1efe70bca8a493ba3a29c7e354c1a45835b4e43c56ca792

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	14 minutes 51 secon...	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
test_rmse	0.403879	14 (15 readings)	explicit
paper_test_rmse	0.403521	14 (15 readings)	explicit
test_rmse_std	0.00282201		explicit

Inline claim	Value	Phase	Output line
test_rmse_mean	0.40378		92
paper_test_rmse_mean	0.403006		93

4 Execution evidence

Coherence

Coherent (0.54): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	85.4 s	2.30 GB	97%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 117 tracked files, 17 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	22:11:28	6cd1493c8b4b8119	73
2	phase seed_1	22:12:31	16cea17e0808e054	3
3	phase seed_2	22:13:54	8ce3ed351ac7a614	3
4	phase seed_3	22:14:53	a7f9141ac1fc4291	3
5	phase seed_4	22:15:58	e69288e8a99b4493	3
6	phase seed_5	22:16:52	e07a5c7066f33608	3
7	phase seed_6	22:17:54	4cc5ab6aac499428	3
8	phase seed_7	22:18:48	025c68fc0158fa04	3
9	phase seed_8	22:19:39	dd72b3b7c5ec884f	3
10	phase seed_9	22:20:26	f9398cf9e8390674	3
11	phase seed_10	22:21:32	a0e135f88c35d3b0	3
12	phase seed_11	22:22:29	561a18b72c005ec1	3
13	phase seed_12	22:23:28	4dc799cfb44e350b	3
14	phase seed_13	22:24:24	35bd2bc01082ef0f	3
15	phase seed_14	22:25:24	786bf014fa1df331	3
16	run_end	22:26:18	c6bef1ace19a8ddc	2

1 snapshot with no file change omitted here; full timeline in Appendix B.

Withheld by .kveritasignore	Size	Hash
notebooks/results.ipynb	<1MB	fa05beadb1524f40e5bf303a

Activity: 75 files read, 149 files written, 31 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T22:26:20.842684+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	95249099ed8bf72d7729dcbb158829162356c4f56f707b1e26bc4a1bdb7f159d
Code bundle hash	6c2a1eef7d58a50bbddb425442af0cc65865e72746dd0ce065b05affcc6745e

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
test_rmse	0.402974	0	61	explicit
paper_test_rmse	0.407119	0	62	explicit
test_rmse	0.404569	1	63	explicit
paper_test_rmse	0.404429	1	64	explicit
test_rmse	0.408171	2	65	explicit
paper_test_rmse	0.403843	2	66	explicit
test_rmse	0.402423	3	67	explicit
paper_test_rmse	0.406475	3	68	explicit
test_rmse	0.406765	4	69	explicit
paper_test_rmse	0.401569	4	70	explicit
test_rmse	0.403704	5	71	explicit
paper_test_rmse	0.400393	5	72	explicit
test_rmse	0.407384	6	73	explicit
paper_test_rmse	0.40379	6	74	explicit
test_rmse	0.404598	7	75	explicit
paper_test_rmse	0.399899	7	76	explicit
test_rmse	0.407541	8	77	explicit
paper_test_rmse	0.405169	8	78	explicit
test_rmse	0.399771	9	79	explicit
paper_test_rmse	0.400561	9	80	explicit
test_rmse	0.401766	10	81	explicit
paper_test_rmse	0.401816	10	82	explicit
test_rmse	0.398955	11	83	explicit
paper_test_rmse	0.401465	11	84	explicit
test_rmse	0.403382	12	85	explicit
paper_test_rmse	0.404289	12	86	explicit
test_rmse	0.400814	13	87	explicit
paper_test_rmse	0.400745	13	88	explicit
test_rmse	0.403879	14	89	explicit
paper_test_rmse	0.403521	14	90	explicit
test_rmse_std	0.00282201		91	explicit

Phase	Line	Time	CPU time	Memory	GPU
seed_0	1	22:11:28	0.4 s	0.44 GB	
seed_1	5	22:12:31	0.0 s	0.02 GB	
seed_2	9	22:13:54	0.0 s	0.03 GB	
seed_3	13	22:14:53	0.1 s	0.02 GB	
seed_4	17	22:15:58	0.0 s	0.02 GB	
seed_5	21	22:16:52	0.0 s	0.02 GB	
seed_6	25	22:17:54	0.0 s	0.02 GB	
seed_7	29	22:18:48	0.1 s	0.02 GB	
seed_8	33	22:19:39	0.1 s	0.02 GB	
seed_9	37	22:20:26	0.1 s	0.02 GB	
seed_10	41	22:21:32	0.1 s	0.02 GB	

Phase	Line	Time	CPU time	Memory	GPU
seed_11	45	22:22:29	0.1 s	0.02 GB	
seed_12	49	22:23:28	0.1 s	0.03 GB	
seed_13	53	22:24:24	0.1 s	0.02 GB	
seed_14	57	22:25:24	0.1 s	0.02 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	22:11:28	6cd1493c8b4b8119	73
1	phase seed_0	22:11:28	6cd1493c8b4b8119	0
2	phase seed_1	22:12:31	16cea17e0808e054	3
3	phase seed_2	22:13:54	8ce3ed351ac7a614	3
4	phase seed_3	22:14:53	a7f9141ac1fc4291	3
5	phase seed_4	22:15:58	e69288e8a99b4493	3
6	phase seed_5	22:16:52	e07a5c7066f33608	3
7	phase seed_6	22:17:54	4cc5ab6aac499428	3
8	phase seed_7	22:18:48	025c68fc0158fa04	3
9	phase seed_8	22:19:39	dd72b3b7c5ec884f	3
10	phase seed_9	22:20:26	f9398cf9e8390674	3
11	phase seed_10	22:21:32	a0e135f88c35d3b0	3
12	phase seed_11	22:22:29	561a18b72c005ec1	3
13	phase seed_12	22:23:28	4dc799cfb44e350b	3
14	phase seed_13	22:24:24	35bd2bc01082ef0f	3
15	phase seed_14	22:25:24	786bf014fa1df331	3
16	run_end	22:26:18	c6bef1ace19a8ddc	2

Event	Path or command	Hash
read	kv_run.sh	
read	exp/tabr/california/0-evaluation/0.toml	
read	.kveritasignore	
read	bin/anp.py	
read	bin/catboost_.py	
read	bin/dkl.py	
read	bin/dnnr.py	
read	bin/dnnr_precompute_scaling.py	
read	bin/ensemble.py	
read	bin/evaluate.py	
read	bin/ffn.py	
read	bin/ft_transformer.py	
read	bin/go.py	
read	bin/knn.py	
read	bin/lightgbm_.py	
read	bin/neighbors.py	
read	bin/saint.py	
read	bin/tabr_add_candidates_after_training.py	
read	bin/tabr_design.py	
read	bin/tabr_scaling.py	
read	bin/tune.py	

Event	Path or command	Hash
read	bin/xgboost_.py	
read	data/california/info.json	
read	environment-simple.yaml	
read	environment.yaml	
read	exp/kv-reproduce/0.toml	
read	exp/tabr/california/0-evaluation/0/report.json	
read	exp/tabr/california/0-evaluation/1.toml	
read	exp/tabr/california/0-evaluation/1/report.json	
read	exp/tabr/california/0-evaluation/10.toml	
read	exp/tabr/california/0-evaluation/10/report.json	
read	exp/tabr/california/0-evaluation/11.toml	
read	exp/tabr/california/0-evaluation/11/report.json	
read	exp/tabr/california/0-evaluation/12.toml	
read	exp/tabr/california/0-evaluation/12/report.json	
read	exp/tabr/california/0-evaluation/13.toml	
read	exp/tabr/california/0-evaluation/13/report.json	
read	exp/tabr/california/0-evaluation/14.toml	
read	exp/tabr/california/0-evaluation/14/report.json	
read	exp/tabr/california/0-evaluation/2.toml	
read	exp/tabr/california/0-evaluation/2/report.json	
read	exp/tabr/california/0-evaluation/3.toml	
read	exp/tabr/california/0-evaluation/3/report.json	
read	exp/tabr/california/0-evaluation/4.toml	
read	exp/tabr/california/0-evaluation/4/report.json	
read	exp/tabr/california/0-evaluation/5.toml	
read	exp/tabr/california/0-evaluation/5/report.json	
read	exp/tabr/california/0-evaluation/6.toml	
read	exp/tabr/california/0-evaluation/6/report.json	
read	exp/tabr/california/0-evaluation/7.toml	
read	exp/tabr/california/0-evaluation/7/report.json	
read	exp/tabr/california/0-evaluation/8.toml	
read	exp/tabr/california/0-evaluation/8/report.json	
read	exp/tabr/california/0-evaluation/9.toml	
read	exp/tabr/california/0-evaluation/9/report.json	
read	lib/__init__.py	
read	lib/anp/module.py	
read	lib/anp/network.py	
read	lib/anp/preprocess.py	
read	lib/deep.py	
read	lib/dnnr/__init__.py	
read	lib/dnnr/dnnr.py	
read	lib/dnnr/nn_index.py	
read	lib/dnnr/scaling.py	
read	lib/env.py	
read	lib/metrics.py	
read	lib/saint.py	
read	lib/util.py	
read	notebooks/results.ipynb	
read	pyproject.toml	
read	setup.cfg	

Event	Path or command	Hash
read	lib/data.py	
read	lib/neighbors.py	
read	bin/tabr.py	
read	cache/build_dataset_california_quantile_None_standard_None_0_38...	
write	exp/kv-reproduce/0/events.out.tfevents.1791411091.localhost.277160.0	d284fc83d9306b66
write	exp/kv-reproduce/0/score_test/events.out.tfevents.1791411092.localhost.2771...	d29bf3b711f51b7d
write	exp/kv-reproduce/0/score_val/events.out.tfevents.1791411092.localhost.2771...	ceb21fdda0f4a9f9
write	exp/kv-reproduce/0/loss_train/events.out.tfevents.1791411092.localhost.2771...	ace4768d3ddccb53
write	exp/kv-reproduce/0/predictions.npz	99eb0df2bed499fe
write	exp/kv-reproduce/0/summary.json	096ae87b143dda92
write	exp/kv-reproduce/0/checkpoint.pt	c563e145280fa460
write	exp/kv-reproduce/0/report.json	0146d37a49b59358
write	exp/kv-reproduce/0/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/1.toml	1ba48c0b80afce3e
write	exp/kv-reproduce/1/events.out.tfevents.1791411154.localhost.279501.0	0c01eb24993f8b09
write	exp/kv-reproduce/1/score_test/events.out.tfevents.1791411155.localhost.2795...	53bd5e4dfa13a736
write	exp/kv-reproduce/1/loss_train/events.out.tfevents.1791411155.localhost.2795...	eac3c4a280011963
write	exp/kv-reproduce/1/score_val/events.out.tfevents.1791411155.localhost.2795...	c36c2a9f2a405ebd
write	exp/kv-reproduce/1/predictions.npz	0722de34ac35b4ef
write	exp/kv-reproduce/1/summary.json	0c036b9d0d55e5d5
write	exp/kv-reproduce/1/checkpoint.pt	99be44725e42b2b6
write	exp/kv-reproduce/1/report.json	33f9cd319b63ecab
write	exp/kv-reproduce/1/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/2.toml	612c1abc8675d19a
write	exp/kv-reproduce/2/events.out.tfevents.1791411237.localhost.282372.0	e74f8b8f9aafe37d
write	exp/kv-reproduce/2/loss_train/events.out.tfevents.1791411238.localhost.2823...	2c8e0718eed09dc4
write	exp/kv-reproduce/2/score_val/events.out.tfevents.1791411238.localhost.2823...	3d1c9af2c05c0025
write	exp/kv-reproduce/2/score_test/events.out.tfevents.1791411238.localhost.2823...	2676c4b3f3bdc800
write	exp/kv-reproduce/2/predictions.npz	a84d699dac6cbdba
write	exp/kv-reproduce/2/summary.json	09a9ff5041988656
write	exp/kv-reproduce/2/checkpoint.pt	76b9538a553b47a3
write	exp/kv-reproduce/2/report.json	ad14f64e5643997c
write	exp/kv-reproduce/2/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/3.toml	a629d354d2ec5a43
write	exp/kv-reproduce/3/events.out.tfevents.1791411296.localhost.284467.0	3a8fe20afbb91e10
write	exp/kv-reproduce/3/score_val/events.out.tfevents.1791411298.localhost.2844...	2dcd667da64495b7
write	exp/kv-reproduce/3/score_test/events.out.tfevents.1791411298.localhost.2844...	eb668324ba351287
write	exp/kv-reproduce/3/loss_train/events.out.tfevents.1791411298.localhost.2844...	c0fbc4a27ec970c2
write	exp/kv-reproduce/3/predictions.npz	36299f9e1a85efc9
write	exp/kv-reproduce/3/summary.json	446eb25afb7c903a
write	exp/kv-reproduce/3/checkpoint.pt	f6cd81a153581aad
write	exp/kv-reproduce/3/report.json	744f8100594afc4c
write	exp/kv-reproduce/3/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/4.toml	ca3351f1498da338
write	exp/kv-reproduce/4/events.out.tfevents.1791411361.localhost.286752.0	4b9c96eaac4d31eb
write	exp/kv-reproduce/4/loss_train/events.out.tfevents.1791411363.localhost.2867...	e8a6a00be3a1b3b9
write	exp/kv-reproduce/4/score_test/events.out.tfevents.1791411363.localhost.2867...	5e63cf0f610ca0be
write	exp/kv-reproduce/4/score_val/events.out.tfevents.1791411363.localhost.2867...	c6b68c9bbfdb1b09
write	exp/kv-reproduce/4/predictions.npz	a83ae9d09cc95f42
write	exp/kv-reproduce/4/summary.json	779d7d5bfd69824f

Event	Path or command	Hash
write	exp/kv-reproduce/4/checkpoint.pt	5b0d56cbb59edb5a
write	exp/kv-reproduce/4/report.json	7c13d5314f7c7fb4
write	exp/kv-reproduce/4/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/5.toml	e72ead8fb6e62810
write	exp/kv-reproduce/5/events.out.tfevents.1791411415.localhost.288626.0	8f2073ce990092ee
write	exp/kv-reproduce/5/loss_train/events.out.tfevents.1791411416.localhost.2886...	8ab499cd4b6c6ceb
write	exp/kv-reproduce/5/score_test/events.out.tfevents.1791411416.localhost.2886...	a43bceb4dc43326
write	exp/kv-reproduce/5/score_val/events.out.tfevents.1791411416.localhost.2886...	cbf68a642f97b460
write	exp/kv-reproduce/5/predictions.npz	b0d6eec0c9319b02
write	exp/kv-reproduce/5/summary.json	1e572c6dad242fa9
write	exp/kv-reproduce/5/checkpoint.pt	574d6096a94ec229
write	exp/kv-reproduce/5/report.json	3ff374f5b3e47a7e
write	exp/kv-reproduce/5/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/6.toml	f97a57ecc180fc9c
write	exp/kv-reproduce/6/events.out.tfevents.1791411477.localhost.290850.0	155f79b32fb22fb0
write	exp/kv-reproduce/6/score_test/events.out.tfevents.1791411479.localhost.2908...	085658f461bdbe59
write	exp/kv-reproduce/6/score_val/events.out.tfevents.1791411479.localhost.2908...	fb808881d54560af
write	exp/kv-reproduce/6/loss_train/events.out.tfevents.1791411479.localhost.2908...	b694d7536304e53a
write	exp/kv-reproduce/6/predictions.npz	8f5e8cf675a47dcc
write	exp/kv-reproduce/6/summary.json	4e80dccb1af64d7
write	exp/kv-reproduce/6/checkpoint.pt	e6eb054cd7377af5
write	exp/kv-reproduce/6/report.json	648ae9f604676cfa
write	exp/kv-reproduce/6/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/7.toml	1e30e72f470fdf62
write	exp/kv-reproduce/7/events.out.tfevents.1791411531.localhost.292754.0	daccb3401c74bf99
write	exp/kv-reproduce/7/score_val/events.out.tfevents.1791411533.localhost.2927...	a146e1d09f51fbf4
write	exp/kv-reproduce/7/score_test/events.out.tfevents.1791411533.localhost.2927...	081c0700e4402794
write	exp/kv-reproduce/7/loss_train/events.out.tfevents.1791411533.localhost.2927...	8d4702b7cb656729
write	exp/kv-reproduce/7/predictions.npz	540332a2662c8f18
write	exp/kv-reproduce/7/summary.json	510cfc1b6c431055
write	exp/kv-reproduce/7/checkpoint.pt	e9e22f2333ae8795
write	exp/kv-reproduce/7/report.json	8975033d6a1e4d63
write	exp/kv-reproduce/7/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/8.toml	d91dbdce69089663
write	exp/kv-reproduce/8/events.out.tfevents.1791411582.localhost.294549.0	6e06fd67a083a4b6
write	exp/kv-reproduce/8/score_test/events.out.tfevents.1791411583.localhost.2945...	da6e21bd74492e8c
write	exp/kv-reproduce/8/loss_train/events.out.tfevents.1791411583.localhost.2945...	74224f931f5783ba
write	exp/kv-reproduce/8/score_val/events.out.tfevents.1791411583.localhost.2945...	00c4e5e72fbd149f
write	exp/kv-reproduce/8/predictions.npz	f0f4e965efa9b9c2
write	exp/kv-reproduce/8/summary.json	5dd99d323082aa6d
write	exp/kv-reproduce/8/checkpoint.pt	e1d6583ffa099ce7
write	exp/kv-reproduce/8/report.json	85174044fd52fcef
write	exp/kv-reproduce/8/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/9.toml	41b56acc8a92f812
write	exp/kv-reproduce/9/events.out.tfevents.1791411629.localhost.296255.0	c744d55e17c18981
write	exp/kv-reproduce/9/score_val/events.out.tfevents.1791411631.localhost.2962...	c97a1f727d66634e
write	exp/kv-reproduce/9/score_test/events.out.tfevents.1791411631.localhost.2962...	c76e6f6c27c9070d
write	exp/kv-reproduce/9/loss_train/events.out.tfevents.1791411631.localhost.2962...	d95a889626b66f18
write	exp/kv-reproduce/9/predictions.npz	c92e11916318a1bf
write	exp/kv-reproduce/9/summary.json	d2d0f8c43d68c9ab

Event	Path or command	Hash
write	exp/kv-reproduce/9/checkpoint.pt	fb5f6b71444fcbd3
write	exp/kv-reproduce/9/report.json	8db84451e5ab1b1d
write	exp/kv-reproduce/9/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/10.toml	831aae329a9f0efc
write	exp/kv-reproduce/10/events.out.tfevents.1791411695.localhost.298585.0	31cbbba610c5871be
write	exp/kv-reproduce/10/score_test/events.out.tfevents.1791411697.localhost.298...	e7b3a7858fdc591b
write	exp/kv-reproduce/10/score_val/events.out.tfevents.1791411697.localhost.298...	1a7ca100088fa36d
write	exp/kv-reproduce/10/loss_train/events.out.tfevents.1791411697.localhost.298...	3256aabecf46d247
write	exp/kv-reproduce/10/predictions.npz	6902663fffcb7980
write	exp/kv-reproduce/10/summary.json	3fc0d2421d431069
write	exp/kv-reproduce/10/checkpoint.pt	0842312f842a3acb
write	exp/kv-reproduce/10/report.json	7e249e3589f4f099
write	exp/kv-reproduce/10/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/11.toml	e9d23e0bba51866d
write	exp/kv-reproduce/11/events.out.tfevents.1791411752.localhost.300600.0	47c37b0c65fbdc3b
write	exp/kv-reproduce/11/score_test/events.out.tfevents.1791411754.localhost.300...	cf36186c5435992b
write	exp/kv-reproduce/11/loss_train/events.out.tfevents.1791411754.localhost.300...	6a3fb92bd0d5af4b
write	exp/kv-reproduce/11/score_val/events.out.tfevents.1791411754.localhost.300...	43b0e5dc84588b39
write	exp/kv-reproduce/11/predictions.npz	ed321a47ccd83e06
write	exp/kv-reproduce/11/summary.json	cae3f5038fffbcb99
write	exp/kv-reproduce/11/checkpoint.pt	056ff865245705d3
write	exp/kv-reproduce/11/report.json	0edd9f2cb7ef95f9
write	exp/kv-reproduce/11/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/12.toml	f441f65e2c68dd19
write	exp/kv-reproduce/12/events.out.tfevents.1791411811.localhost.302697.0	fb5d50d49e0aed8e
write	exp/kv-reproduce/12/loss_train/events.out.tfevents.1791411813.localhost.302...	95cdf6d5a8806b8c
write	exp/kv-reproduce/12/score_test/events.out.tfevents.1791411813.localhost.302...	404d7e746af19d84
write	exp/kv-reproduce/12/score_val/events.out.tfevents.1791411813.localhost.302...	163d373ec9704038
write	exp/kv-reproduce/12/predictions.npz	336bd27ea8edeb8b
write	exp/kv-reproduce/12/summary.json	3fbc8ccad5afea90
write	exp/kv-reproduce/12/checkpoint.pt	adf7f92805586989
write	exp/kv-reproduce/12/report.json	5445f2f7fcee701
write	exp/kv-reproduce/12/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/13.toml	10558015f9725a74
write	exp/kv-reproduce/13/events.out.tfevents.1791411867.localhost.304655.0	9358e7daf90a52c8
write	exp/kv-reproduce/13/score_val/events.out.tfevents.1791411869.localhost.304...	62d53d2ab7e5687a
write	exp/kv-reproduce/13/loss_train/events.out.tfevents.1791411869.localhost.304...	00badc7c7b641e80
write	exp/kv-reproduce/13/score_test/events.out.tfevents.1791411869.localhost.304...	0a426faeb4e4d72d
write	exp/kv-reproduce/13/predictions.npz	465f3a9771aab2e2
write	exp/kv-reproduce/13/summary.json	ea0db9787fba291b
write	exp/kv-reproduce/13/checkpoint.pt	77214a66333b736d
write	exp/kv-reproduce/13/report.json	d1e63a79da0cb090
write	exp/kv-reproduce/13/DONE	e3b0c44298fc1c14
write	exp/kv-reproduce/14.toml	40eaaafab460f62e9
write	exp/kv-reproduce/14/events.out.tfevents.1791411927.localhost.306786.0	3504b660a93bc1d5
write	exp/kv-reproduce/14/score_test/events.out.tfevents.1791411929.localhost.306...	ffc37275e98aacf3
write	exp/kv-reproduce/14/score_val/events.out.tfevents.1791411929.localhost.306...	4dbab016a73a2f07
write	exp/kv-reproduce/14/loss_train/events.out.tfevents.1791411929.localhost.306...	9778f7de4df1d711
write	exp/kv-reproduce/14/predictions.npz	77f5051856b04142
write	exp/kv-reproduce/14/summary.json	2a38161e55a25c0b

Event	Path or command	Hash
write	exp/kv-reproduce/14/checkpoint.pt	9f811a0efee8ae73
write	exp/kv-reproduce/14/report.json	1de4407d64d0e8a3
write	exp/kv-reproduce/14/DONE	e3b0c44298fc1c14
process	python bin/tabr.py exp/kv-reproduce/0.toml --force	
process	tail -3	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/1.toml --force	
process	python bin/tabr.py exp/kv-reproduce/2.toml --force	
process	tail -3	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/3.toml --force	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/4.toml --force	
process	python bin/tabr.py exp/kv-reproduce/5.toml --force	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/6.toml --force	
process	tail -3	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/7.toml --force	
process	python bin/tabr.py exp/kv-reproduce/8.toml --force	
process	tail -3	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/9.toml --force	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/10.toml --force	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/11.toml --force	
process	python bin/tabr.py exp/kv-reproduce/12.toml --force	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/13.toml --force	
process	tail -3	
process	python bin/tabr.py exp/kv-reproduce/14.toml --force	
process	tail -3	
process	python - exp/tabr/california/0-evaluation exp/kv-reproduce	

C Run history

Invocation 1, OK, 14 minutes 51 seconds, 93 output lines, started 2026-10-07T22:11:27

metric_hash 53e6f7e72a91ec0eb3b2178e219a2dab7878e66716f01d7d24948f8d45e22fbe
run_digest 5c05756341947dee7aa6c03a4d182c8132b4d15625a3557b0c0be01773241a6c
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 95249099ed8bf72d7729dcbb158829162356c4f56f707b1e26bc4a1bdb7f159d
signed_message_hash d227f232dcc7de744c25450d8a109f48c468487ca67cf56906f3a393e39a4790
nonce 8de724257a0a8b08f1126f2ae6f77195
signed_at 2026-10-07T22:26:20.842684+00:00
sealed_at 2026-10-07T22:26:20.903492162Z
checkout_bundle 6c2a1eef7d58a50bbddb425442af0cc65865e72746dd0ce065b05affcc6745e

Signature (base64)

ZxpFIIB+UBFbfkhJa4NFnx1DyivEmxFMjbhoUUTcxzzBAKqeXby5aDQ6byxJXzsDSvcFnA/3PC7kJLtrMXA47F94
 X25DLQYDGIvbfV2cKLiI/MjjL3CvfU1lUka1ezSa0DcGC7WwkJHKCby4dEDzEeE+0aKsFRowYzgCqwsMwk5aCHcA
 IBn0pyNZrJWjhpWghACJuY7bADB934z8vkDjmyT4k2sKT7JXVRFKBKV/K9PDCacLhtcugDQAB2Di0o+bsial40jN
 isZ67P4HFjqw1w0XmGZfN6mcn/1J6v8tA6TQm4tb0aaB+qcULfX03n0BohKLZsKuJQIPRUF80Ku5NyozJnqZhg/8
 +RBM9uHvcJssIMziFECNExjWiiXx63b+jahGWrXWvWI7CokmgYCBypckKgtJ0VLp+uM5C/6s1V/nIu+ldyFzh0HR
 +MfyloAaCdDUKlallwVvt/pw4wu0g/DuJGY57zQEg0ir828oCCTN0d0hv5QE+U4EZ0lbh9da0oVhU0N1wckndVBMB
 HpZdFJuauOSNADfbwyGUJRsr7Tec1uX8NDMDkAGtZOAXwx2R2Jw++3gXggDDExnHAw99TmzL/S2dTya8SLVAoEIJ
 qLGWYTI964wW8HbXr08ICS16LwGmHgSIG0fa2duFG802j6ae7cNAXmiku+2YOU8aZns=

Public key

-----BEGIN PUBLIC KEY-----

MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAlQL3YyMgeShx8IhTC7jb
 sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LoUFz5CUZzRi
 6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
 7qu95EFEEisX/GfoS9ToXtqPBwCcVwSPBX8efec4TC58m96Fsp4mC8KIcEadbIbHN
 av0JjdTJ//CMOWZSIbdzE6qvEbNfcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
 AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqeS7VSzt3o
 XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
 HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNcy9bLgPU5annzP9gN2l
 6HEMLdkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
 1GN7LE3mbQRbPsLL/JapDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
 +eE4bf/olCHBCf1VIDCkDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
 8Jap/1PwjNcNku5cdW3gHLCcAwEAAQ==

-----END PUBLIC KEY-----

Run 1 outputs

```
stdout a943b92c6cd28dda1af11035334e8b2f088ec3743e2c7a2fd3c72d9a82dce824
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 53e6f7e72a91ec0eb3b2178e219a2dab7878e66716f01d7d24948f8d45e22fbe
digest 5c05756341947dee7aa6c03a4d182c8132b4d15625a3557b0c0be01773241a6c
kv_run.sh (unchanged)
pre e1b8d68478065b21e96b8d4252f4b9ce71f75ff4211719d485dcbc8af5b68983
post e1b8d68478065b21e96b8d4252f4b9ce71f75ff4211719d485dcbc8af5b68983
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```